

Mode 5

$$\text{DP32 } (8 \times 4) \rightarrow \text{DP16 } (8 \times 4) \rightarrow \text{DP8 } (8 \times 4)$$

Step 1 — DP32 (8×4)

$$\text{DP32 } (8 \times 4) = \sum_{i=0}^{31} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^{15} a_i b_i}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=16}^{31} a_i b_i}_{\text{DP16 } (8 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=16}^{23} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=24}^{31} a_i b_i}_{\text{DP8 } (8 \times 4)}$$